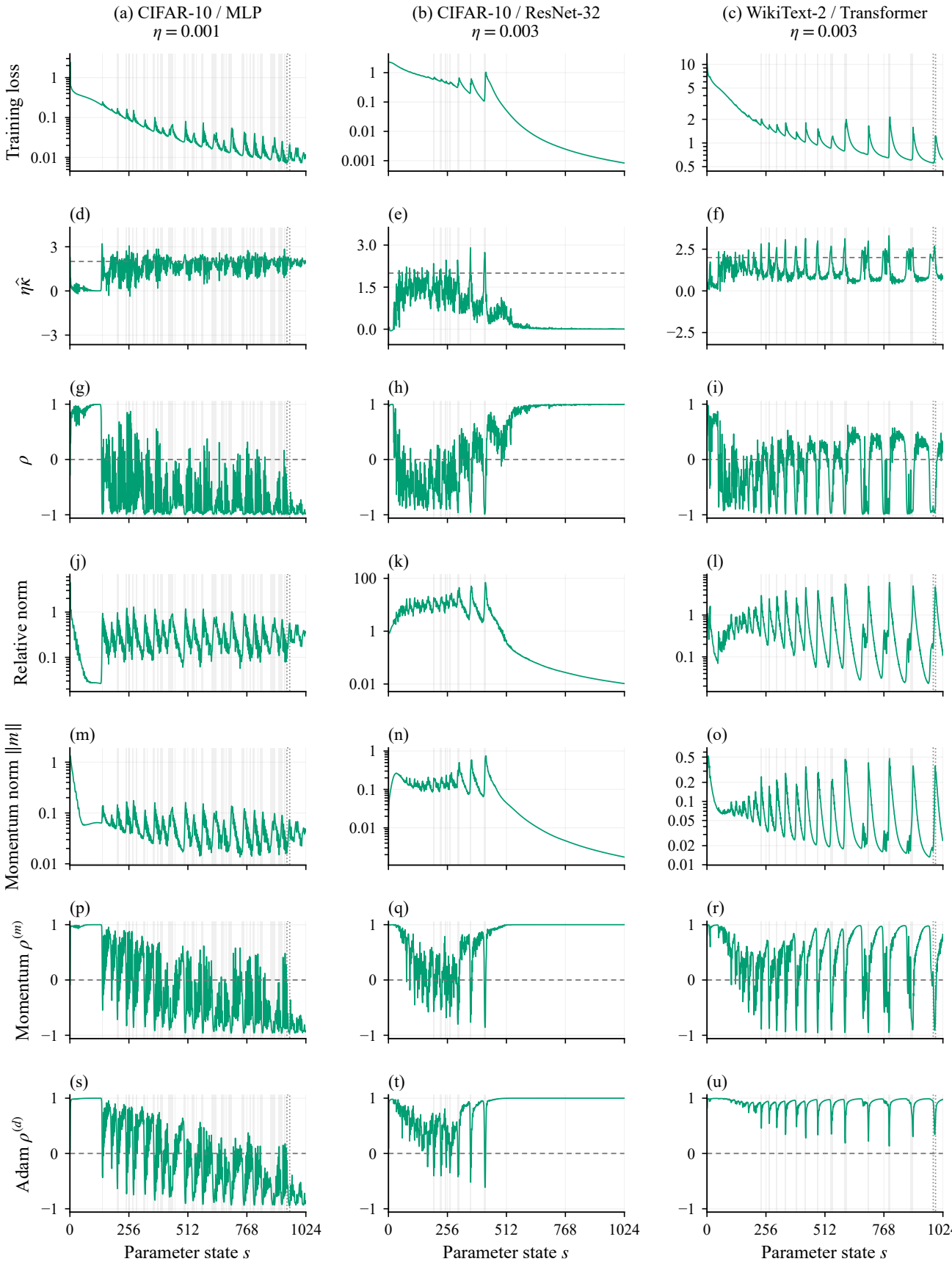

(a) CIFAR-10 / MLP
$\eta = 0.001$
(b) CIFAR-10 / ResNet-32
$\eta = 0.003$
(c) WikiText-2 / Transformer
$\eta = 0.003$
Training loss
(d)
(e)
(f)
$\hat{\eta\kappa}$
(g)
(h)
(i)
ρ
(j)
(k)
(l)
Relative norm
(m)
(n)
(o)
Momentum norm $\|m\|$
(p)
(q)
(r)
Momentum $\rho^{(m)}$
(s)
(t)
(u)
Adam $\rho^{(d)}$
0
256
512
768
1024
Parameter state s
Rise of recovered >10% excursion
Recovery unobserved